

Jorge Becerra Garrido

Curriculum Vitae¹

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RESEARCH INTERESTS

I am mostly interested in quantum topology. The following is a non-exhaustive list of my interests:

- Quantum knot invariants, especially those which arise from ribbon categories/algebras.
- Tangle categories.
- Finite type invariants and the Kontsevich invariant.
- Modular categories, modular functors, modular operads.
- Factorisation homology, skein categories.

EMPLOYMENT

Université Bourgogne Europe (France). 2024 – present
Postdoc in the research group of L. Woike.

EDUCATION

Ph.D. in Mathematics, *University of Groningen* (The Netherlands). 2019 – 2024
Thesis: *Universal quantum knots invariants*, supervised by R. van der Veen.

M.Sc. Mathematical Sciences, *Utrecht University* (The Netherlands). 2017 – 2019
Honours programme UGC, Cum Laude.
Thesis: *K-theory with Reality*, supervised by L. Meier.

B.Sc. Mathematics, *University of Extremadura* (Spain). 2013 – 2017
Thesis: *Theory of Relativity*, supervised by J.A. Navarro.

PUBLICATIONS & PREPRINTS

1. *Minimal generating sets of rotational Reidemeister moves* (preprint).
2. *A refined functorial universal tangle invariant* (preprint).
3. *Strictification and non-strictification of monoidal categories* (preprint).
4. *On Bar-Natan - van der Veen's perturbed Gaussians*, Rev. Real Acad. Cienc. Exactas Fis. Nat. Ser. A-Mat., 118(2):46, 2024.
5. *A combinatorial PROP for bialgebras*. European J. Combin. 124 (2025), Paper No. 104086.
6. *An Introduction to Knot Homology Theories*, TEMat monográficos, 2 (2021): Proceedings of the 3rd BYMAT Conference, pp. 11-14. ISSN: 2660-6003.
7. *Finite, Infinite and Twisted Ascent and Descent* (joint with JMF Castillo), La Gaceta de la Real Sociedad Matemática Española, Vol. 21 (2018), Núm. 3, Págs 497–516.

¹Last updated: 3 July 2025

RESEARCH TALKS

- *Hopf-algebraic quantum knot invariants*, [GADT groupe de travail](#), Dijon. Nov 2024
- *Universal quantum knot invariants*, [Groningen Quantum Topology Dag](#). May 2024
- *Hopf algebras and knot invariants*, [ICMAT Geometry Seminar](#), Madrid. Jan 2024
- *Drinfeld doubles and Gaussian calculus for universal knot invariants*, [MAPHYAG Seminar](#), Madrid. Jan 2024
- *Quantum invariants of knots and tangles*, [IMUEX Seminar](#), Badajoz. Jan 2024
- *Hopf algebras and the 2-loop polynomial of knots*, [Algebra Seminar](#), Groningen. Dec 2022
- *Three faces of the 2-loop polynomial of knots*, [4th BYMAT](#), online. Nov 2022
- *Universal quantum knot invariants*, [GQT School & Colloquium](#), Zeist. Aug 2022
- *Quantum invariants via perturbed Gaussian expressions*, [Winter Braids XI](#), Dijon Dec 2021
- *A categorical approach to the universal tangle invariant*, [DIAMANT symposium](#), Utrecht. Nov 2021
- *An introduction to knot homology theories*, [3rd BYMAT](#), online. Dec 2020
- *Foundations of stable homotopy theory*, [IMUEX Seminar](#), Badajoz. May 2019
- *Functional inequalities and convergence in diffusion processes*, [13th Meeting on Functional Analysis and Applications](#). Cáceres. Mar 2017

For a list of non-research talks, see sites.google.com/view/becerra/talks.

SELECTED ATTENDANCE

- [Matemale spring school](#), Matemale. May 2025
- [Algebraic, geometric and quantum topology in Picardie II](#), Amiens. Mar 2025
- [Skein days](#), Cergy-Pontoise. Dec 2024
- [Algebraic, geometric and quantum topology in Picardie](#), Amiens. Mar 2024
- [Swiss Knots 2023](#), Regensburg. Sep 2023
- [From Subfactors to Quantum Topology - In memory of Vaughan Jones](#), Geneva. Jun 2022
- [Quantum Topology and Geometry \(in honor of Vladimir Turaev\)](#), Paris Jun 2022
- [Winter Braids XI](#), Dijon. Dec 2021
- [Georgia Tech Topology Summer School on 4-manifolds](#), Online. Jul 2021
- [Monoidal and 2-categories in representation theory and categorification](#), Online Dec 2020
- [QUAntum groups, Categorification, Knot invariants, and Soergel bimodules](#), Online. Aug 2020

For a complete list, see sites.google.com/view/becerra/attendance.

TEACHING

- **University of Groningen:**
 - Co-lecturer of [Topics in Topology](#) (Master, universal knot invariants). Spring 2023
 - Co-lecturer of [Topics in Topology](#) (Master, Kirby calculus). Spring 2022
 - Co-lecturer of [Topics in Topology](#) (Master, grid homology). Spring 2021
 - Teaching Assistant for [Multivariable Analysis](#) (Bachelor). Autumn 2020
 - Co-organiser of the [Knot Theory Seminar](#) (Bach./Master). Spring 2020
 - Co-organiser of the [Groningen Topology Seminar](#) (Bach./Master). Autumn 2019

- **Utrecht University:**

Teaching Assistant for <i>Topologie en Meetkunde</i> (Bachelor, π_1).	Spring 2019
Teaching Assistant for <i>Algebraic Topology</i> (Master, H_n 's and π_n 's).	Autumn 2018
Teaching Assistant for <i>Topologie en Meetkunde</i> (Bachelor, π_1).	Spring 2018

BACHELOR'S THESES SUPERVISION

- Daniël Apol, *Representability of cohomology theories on CW-complexes*, Groningen. 2021

SKILLS

- **Languages:** Spanish (native), English (fluent), French (intermediate), Dutch (basic).
- **Programming languages:**
Python (with [verified certificate](#) by HarvardX), including SAGE, Numpy, Pandas, Matplotlib, NetworkX and Scikit-learn.
Mathematica, C, R-Commander, Fortran, L^AT_EX, Linux.
- **Teaching:** Teacher training certificate.

GRANTS

- [Research Assistant](#), Department of Mathematics, University of Extremadura (Badajoz, Spain). €2,000. 2016–2017
- Fee waiver for talented students, *Summer School on Geometry*, Utrecht (The Netherlands). €400 2016
- Twelve university course tuition fee waivers for top student of a course. €1,200 2013–2016
- One-year university tuition fee waiver for top high-school student. €1,000 2013

ORGANISATION

- Co-organiser of the workshop [Groningen Quantum Topology Dag](#) May 2024

PROFESSIONAL ACTIVITIES

- Organiser of a number of seminars and reading groups at Groningen University, cf. [PhD Seminar](#), [Knot homology theories reading group](#), [Quantum Mechanics bookclub](#).
- Speaker at the *European Researchers' Night* and similar outreach activities at University of Extremadura (2014 – 2017).
- Private tutor of English, Mathematics and Physics for high school students (2013 – 2017).